LEC09 - Discrete Random Variables

Discrete Random Variable

- A random variable is said to be discrete if it can assume only a finite or countably infinite number of distinct values
- Ex. If X represents the outcome of a six-sided die, it can only take on values 1,2,3,4,5,6

Probability Distributions

- Capital letters denote random variables (X, Y, Z)
- Small letters denote values of a random variable (x, y, z)
- Write $P(X=x)$ which means "Probability that X takes on x"
- Every random variable is associated with a probability distribution which specifies random variable values and the probability each value will occur
- The probability distribution for a discrete random variable X can be represented by a formula, table, or graph that provides $P(X=x)$ for all $x \in D$
- Note that $P(X=x) \geq 0$ for all $x \in D$, but the probability distribution only assigns nonzero probabilities to a countable number of x values
- It is convention that any value x not explicitly assigned a positive probability is 0 $(P(X=x)=0)$
- Probability distribution of the random variable X tells us how the probability of 1 is distributed among all values of X

Example

Let the random variable X be the sum of a pair of dice: $P(X=x) = \frac{6-|x-7|}{36}$ for $x = 2, 3, \cdots, 12$, zero otherwise

x	2	3	4	5	6	7	8	9	10	11	12
$P(X=x)$	$1/36$	$2/36$	$3/36$	$4/36$	$5/36$	$6/36$	$5/36$	$4/36$	$3/36$	$2/36$	$1/36$

Probability Mass Functions

- For discrete random variables, the probability distribution is called the probability mass function (pmf) and is denoted by

$$p_X(x) = P(X=x),$$

  for $x \in D$
- We often remove the subscript x $(p_X(x) = p(x))$
- For any discrete probability distribution, all the following must be true:

$$\left.\begin{array}{l} 1.\ 0 \leq p(x) \leq 1 \text{ for all } x \\ 2.\ \sum_x p(x) = 1 \text{, where the summation is over all values of } x \text{ with nonzero probability} \end{array}\right\} \text{Probability axioms applied to random variables}$$

- The induced probability distribution, $p_X(.)$ of X is

$$p_X(.) = \sum_{d_i \in Y} p(d_i), \quad Y \subset D$$

Example

A recruiting manager has 6 internship applications, three men and three women. He has chose to select two at random. Let X denote the number of women. Find the probability distribution for X.

$$p(x) = \frac{\binom{3}{x}\binom{3}{2-x}}{\binom{6}{2}}, \text{ for } x = 0, 1, 2$$

Cumulative Distribution Functions

- Let X denote any random variable. The cumulative distribution function (cdf) of X denoted by $F_X(x)$, is defined by

  $$F_X(x) = P(X \le x), \text{ for } -\infty < x < \infty$$

- We often remove the subscript X ($F_X(x) = F(x)$)
- If $F(x)$ is a cdf, then:
  1. $F(-\infty) = \lim_{x \to -\infty} P(X \le x) = \lim_{x \to -\infty} F(x) = 0$
  2. $F(\infty) = \lim_{x \to \infty} P(X \le x) = \lim_{x \to \infty} F(x) = 1$
  3. $F(x)$ is a right non-decreasing function of x
     - $F(x_1) \le F(x_2)$ for any $x_1 < x_2$
- Difference between cdf of continuous and discrete random variables
- The cdf for a discrete random variable is a step function
  - Jump at each value of x which has positive probability, flat otherwise
  - Occurs because cdf only increases at finite or countable numbers of points with positive probabilities